COMPUTER GRAPHICS — TERM PROJECT PROPOSAL

A Resolution-Independent Vulkan Fluid Brush for Procedural Marble at 1080p and ≤16.6 ms per GPU Update

1. TEAM (INDIVIDUAL)

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Role: Solo developer responsible for Vulkan/GPU implementation, brush algorithm, procedural marble, testing and evaluation, demo, and documentation.

2. PROBLEM

We will build an interactive 2D painting prototype for students and graphics learners to explore how a brush can mix color across a procedural marble surface. The challenge is to make the brush update GPU-first, responsive, and measurable at practical output resolutions. The target platform is a desktop with a Vulkan 1.3+

3. RESEARCH ANCHOR

Research anchor: Mixwell, “Sharp 2D Fluid Brushes for Progressive Physics-Based Mixing” (2026). Official project page:
https://dougjam.github.io/mixwell-2026/

We will inspect the paper and official supplementary materials/code, then implement and clearly identify one feasible brush-mixing idea that is verified in those sources. We will not claim to reproduce the complete paper. Our project contribution is the Vulkan integration, procedural marble field, interactive controls, and repeatable evaluation.

Out of scope: a full fluid simulation, a general-purpose painting package, layers, multiple brush
families, and a complete paper reproduction.

4. SUCCESS CRITERIA

Measure GPU brush-update time with Vulkan timestamps at 1920×1080, using a fixed stroke workload, after 30 warm-up updates and over 200 measured updates. Report the median and spread; report results only on the recorded test GPU.

Criterion	Minimum	Stretch
Median GPU brush update, 1080p	≤16.6 ms	≤8.0 ms
GPU vs. deterministic CPU reference	PSNR ≥35 dB in the affected brush region	PSNR ≥40 dB in the affected brush region
Repeatability	Same fixed seed, initial canvas, and stroke path	Same, with results reproduced in a second run

5. DESIGN

Pipeline:

Mouse/stroke input ⟶ CPU validates and packs brush parameters and stroke segments ⟶ GPU evaluates the deterministic procedural marble field and applies the selected brush-mixing update to the affected canvas region ⟶ GPU renders the canvas ⟶ Vulkan timestamp queries capture GPU pass times.

GPU resources: canvas color storage/sampled image; storage image for the updated result if required by the chosen ping-pong design; compact stroke segment storage buffer; marble/brush parameter uniform or push-constant data; descriptor sets; timestamp query pool. Shader/host layouts and image synchronization will be checked against the implementation.

6. EVALUATION

Baseline: a simple GPU color-blend brush on the same procedural marble, compared with the research-informed mixing update using identical initial canvas, stroke path, brush settings, test GPU, and resolution. Report each GPU pass time, end-to-end GPU brush-update time, memory/resolution, and GPU-versus-CPU-reference PSNR. Define PSNR preprocessing, color space, and affected-region mask consistently before collecting results.

Ablations:

A. Output resolution: 512×512, 1920×1080, and 2560×1440; keep the procedural domain, feature scale, and mapped brush path consistent.

B. Stroke workload: 1, 4, and 8 segments per update; keep resolution and brush settings fixed.

7. RISKS & PLAN B

Risk: the paper's details, supplementary material, or license may limit what can be verified or reused.

Plan B: implement only a source-verified, feasible subset, cite it accurately, and ask before substituting another method.

Risk: performance or quality targets may be missed or timestamp support may be limited.

Plan B: reduce the supported workload/resolution, retain GPU timestamps where available, document any measurement limitation, and report actual results rather than fabricated values.

8. AI USE & ETHICS

AI assistance will be used for drafting, implementation support, and debugging.

I will verify technical claims against the cited paper/official materials, review and test generated code, and keep an AI-use log describing the assistance, affected files, and checks. No external artwork or dataset is planned; marble is generated procedurally. Any later external source, asset, or dataset will be cited and used only under a verified compatible license.

9. WEEKLY PLAN (W1–W5)

W1 - Finalize scope, verify the research source, submit this proposal, and provide a runnable Vulkan skeleton in the repository.

W2 - Set up Vulkan resources and rendering; prototype the deterministic marble field and source-verified brush method; prepare the CPU reference and benchmark plan.

W3 - Integrate the GPU brush, validate synchronization and correctness, run the midpoint demo, and record initial results.

W4 - Profile performance, complete algorithm tests, run the baseline and both ablations, and prepare reproducibility instructions and source records.

W5 - Complete the demo/page and final ACM report, and verify a clean build.